

William Jackson

Senior Software Engineer – C#/.NET Azure/AWS/GCP

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience designing and scaling **backend platforms** for **SaaS, enterprise workflow, and cloud-based business systems** using **C# (.NET 6–8), ASP.NET Core 6–8**, and distributed service design. Strong background in building secure **REST APIs**, modernizing legacy **.NET Framework** applications into cloud-ready services, and improving reliability through observability, performance tuning, and automated delivery. Known for solving difficult production issues, reducing latency, stabilizing releases, and delivering backend features that support high-volume integrations, data processing, authentication, and multi-environment deployments across **AWS, Azure**, and containerized infrastructure.

SKILLS

- **Backend: C# (.NET 6–8), ASP.NET Core 6–8**, .NET Framework, Web API, Minimal APIs, background services, microservices, distributed systems, multitenant backend architecture
- **Data & Persistence: Entity Framework Core 6–8**, Dapper, SQL Server, PostgreSQL, MySQL, Redis, MongoDB, DynamoDB, query optimization, indexing, transaction handling, caching strategies
- **Messaging & Integration: RabbitMQ, Kafka, AWS SQS/SNS, EventBridge**, webhook integrations, third-party API integration, asynchronous processing, event-driven workflows
- **Cloud & DevOps: AWS, Azure, EC2, ECS, EKS, Lambda, API Gateway, S3, RDS, CloudWatch, Docker, Kubernetes, Helm, Terraform, GitHub Actions, Azure DevOps, CI/CD pipelines**
- **Observability & Security: OpenTelemetry, Datadog, Prometheus, Grafana**, structured logging, health checks, tracing, OAuth2, OpenID Connect, JWT, RBAC, audit logging, rate limiting
- **Testing & Quality: xUnit, NUnit, Moq**, integration testing, API contract validation, performance diagnostics, code reviews, refactoring, release hardening

WORK HISTORY

Senior Software Engineer

February 2021 to December 2025

Viario Networks Inc. - Delaware, United States

- Architected and delivered **backend services** for a cloud-hosted SaaS platform using **C# (.NET 6–8), ASP.NET Core, SQL Server, Redis**, and **RabbitMQ**, creating stable service boundaries that supported high-volume transactional workflows and partner integrations.
- Led a modernization effort that migrated core business modules from a legacy **.NET Framework** monolith into containerized **ASP.NET Core** services, which improved deployment flexibility and reduced release coordination overhead across environments.
- Designed secure **REST APIs** for account provisioning, billing workflows, reporting pipelines, and internal administration capabilities, keeping service contracts consistent so frontend and integration teams could move faster with fewer rework cycles.
- Resolved a recurring production issue where long-running database calls caused timeout spikes under parallel load, then introduced query tuning, targeted indexes, and background offloading that reduced peak API response time by **38%**.
- Implemented centralized **JWT**-based authentication and **RBAC** rules across internal and external service endpoints, which improved authorization consistency and reduced role leakage defects found during release validation by **27%**.
- Built asynchronous processing pipelines with **RabbitMQ** and scheduled workers for notification, reconciliation, and retry-heavy integration flows, which increased completion reliability for delayed tasks by **31%** during high-traffic periods.
- Investigated a difficult memory growth problem in one backend service caused by unmanaged serialization patterns and oversized cached objects, then reworked lifecycle handling and cache boundaries to stabilize container memory usage in production.
- Introduced **OpenTelemetry**, structured logging, and dashboard-driven monitoring with **Datadog** and **Grafana**, allowing the team to trace cross-service failures faster and shorten mean time to resolution by **42%**.

- Delivered a new audit logging capability for sensitive business actions, giving support and compliance stakeholders clearer event visibility while preserving request performance through asynchronous write strategies.
- Improved deployment confidence by building **CI/CD** pipelines in **GitHub Actions** and **Azure DevOps**, enforcing test gates, artifact versioning, and safer rollout controls across lower and production environments.
- Worked closely with product and QA teams to break down backend requirements into incremental deliverables, which helped reduce feature rollback frequency and improved sprint acceptance consistency for service-heavy releases.
- Supported flexible cross-team ownership by moving between API design, database tuning, cloud debugging, incident response, and release hardening whenever delivery risk appeared, making backend execution more predictable during priority shifts.

Software Engineer
Avantia, Inc. - Ohio, United States

October 2018 to January 2021

- Built and maintained business-critical backend applications using **C#, ASP.NET Core, Entity Framework Core, SQL Server**, and **Azure**, supporting internal operations, reporting, and customer-facing process automation.
- Enhanced several legacy service areas by refactoring tightly coupled repository and service layers into cleaner domain-oriented modules, making future feature implementation safer and improving maintainability for shared backend code.
- Delivered secure API endpoints for user lifecycle management, document processing, and operational reporting, ensuring that validation, authorization, and error-handling behavior remained consistent across service boundaries.
- Solved a frustrating defect where concurrent updates occasionally produced stale writes and duplicate records, then introduced transaction controls and optimistic concurrency handling that reduced related support incidents by **34%**.
- Implemented background job processing for exports, notifications, and long-running reconciliation tasks so that user-facing requests stayed responsive even when backend operations became compute or I/O intensive.
- Participated in a staged migration from older .NET application patterns into newer **ASP.NET Core** service design, helping the team preserve business rules while upgrading deployment structure and dependency management.
- Improved SQL performance on heavily filtered reporting endpoints through execution plan analysis, index refinement, and selective query rewrites, cutting average report generation time by **41%** in the most-used workflows.
- Introduced environment-aware configuration management and secrets handling for deployment pipelines, reducing manual configuration mistakes and improving release consistency across development, QA, and production.
- Added a new integration feature that synchronized external customer records with internal workflow services, including retry logic, validation rules, and traceable error states to simplify operational support.
- Strengthened test coverage with **xUnit**, integration tests, and mock-based service validation, which helped catch regression-prone changes earlier and improved pre-release confidence by **29%** across backend tickets.
- Contributed beyond assigned scope by helping with release troubleshooting, production diagnostics, and API contract alignment, showing the flexibility to support architecture, delivery, and incident resolution needs in the same cycle.

Software Developer
ArnAmy, Inc. - Florida, United States

March 2016 to September 2018

- Developed backend modules for internal enterprise systems using **C#, .NET, ASP.NET Web API, SQL Server**, and service-oriented patterns that supported account workflows, approval processes, and operational reporting.
- Maintained and extended legacy application code while gradually introducing cleaner service abstractions, giving the team a more practical path to modernize without interrupting active business operations.

- Built API endpoints and backend logic for form submission, user permissions, data export, and process automation, focusing on reliable request handling and predictable validation behavior.
- Tracked and fixed a recurring null-handling issue that surfaced only during specific integration edge cases, then added stronger defensive mapping and validation logic that reduced related production errors by **24%**.
- Worked on a backend migration initiative that moved selected modules away from older tightly bound application layers into reusable service components, improving code reuse and preparing the platform for later cloud adoption.
- Improved slow SQL queries used by large internal reports through stored procedure tuning, indexing changes, and result-shaping adjustments, which reduced heavy report execution time by **33%** for business users.
- Created support tooling and log enrichment patterns that made production troubleshooting easier for senior engineers, especially during API failures and scheduled job exceptions.
- Helped implement a new backend feature for automated notifications and approval-state transitions, reducing manual follow-up work and improving response flow consistency across departments.
- Collaborated with QA and business analysts to reproduce defects, verify edge cases, and stabilize releases, which improved issue closure quality and reduced back-and-forth during validation cycles.
- Adapted comfortably between maintenance work, new feature delivery, bug investigation, and partial modernization efforts, building the versatile backend discipline needed for larger distributed platform work later in the career.

Junior Software Developer
Centric Tech Inc. - New Jersey, United States

October 2014 to February 2016

- Developed backend modules for internal enterprise systems using **C#, .NET, ASP.NET Web API, SQL Server**, and service-oriented patterns that supported account workflows, approval processes, and operational reporting.
- Maintained and extended legacy application code while gradually introducing cleaner service abstractions, giving the team a more practical path to modernize without interrupting active business operations.
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EDUCATION

Bachelor's degree in Computer Science
Harvard University

2010 to 2014

- Built strong foundations in **algorithms, databases, and software engineering**, applying coursework through practical team projects and production-style system design.